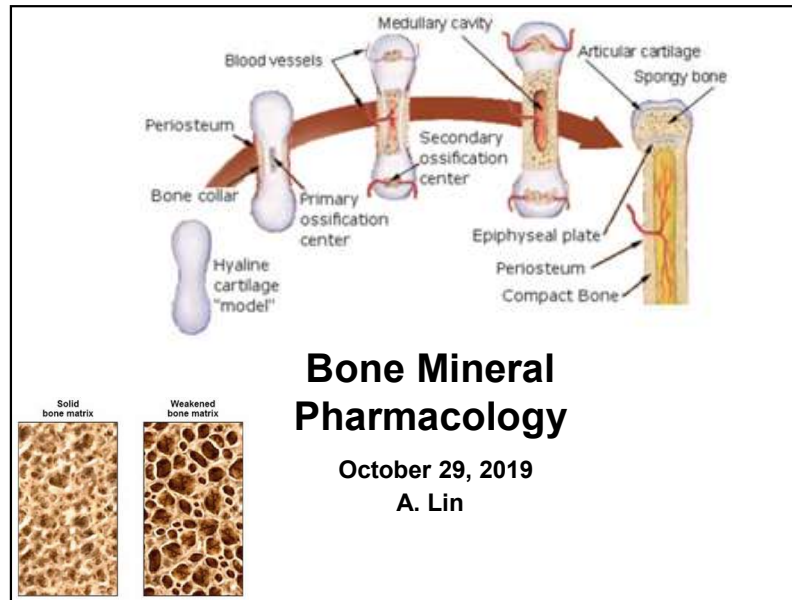




Bone Mineral Pharmacology

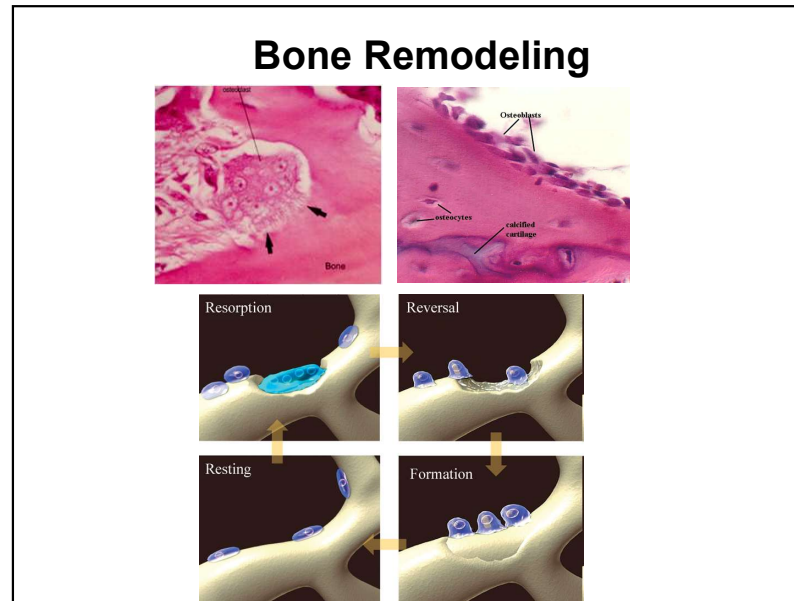
Objectives

- Elucidate the **MOA** of hormones and drugs that regulate bone minerals
- Define the **effect** of PTH and Vit D on **calcium** and **phosphate** (through their actions on kidney and bone)
- Elucidate the **metabolism** of Vit D
- Summarize the **major side effects** of glucocorticoids on bone metabolism
- List the drugs used to treat **osteoporosis** and their **MOA**
- Summarize the **major side effects/caution** in the use of BPs and denosumab

Readiness Preparation

- [Bone Biology](#) (2:34)
- [Bone remodeling](#) (4:13)
- [OPG & RANK/RANKL Pathway](#) (1:54)
- [Osteoporosis](#)

Sign up for Medscape



Normal bone remodeling involves four major processes:

- (i) **Resorption:** stimulated osteoblast precursors release factors that induce osteoclast differentiation and activity. Osteoclasts remove bone mineral and matrix, creating an erosion cavity.
- (ii) **Reversal:** mononuclear cells prepare bone surface for new osteoblasts to begin forming bone.
- (iii) **Formation:** successive waves of osteoblasts synthesize an organic matrix to replace resorbed bone and fill the cavity with new bone.
- (iv) **Resting:** bone surface is covered with flattened lining cells. A prolonged resting period follows with little cellular activity until a new remodeling cycle begins



- ❑ The treatment of osteoporosis consists of **lifestyle measures** and **pharmacologic therapy**.
- ❑ **Lifestyle measures include**
 - adequate calcium and vitamin D
 - exercise
 - smoking cessation
 - counseling on fall prevention
 - avoidance of heavy alcohol use

Hormones and Drugs that Affect Bone Mineral Homeostasis

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	Vit D & analogs
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	Estrogens
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Glucocorticoid: negative effect on bone health (long term use leads to osteoporosis)

Specific Disorders Involving the Bone Mineral-Regulating Hormones

- Primary Hyperparathyroidism
- Hypoparathyroidism
- Nutritional Vitamin D Deficiency or Insufficiency
- Chronic Kidney Disease
- Intestinal osteodystrophy
- **Osteoporosis**
- X-linked & Autosomal Dominant Hypophosphatemia
- Vitamin D-Dependent Rickets Types I & II
- Nephrotic Syndrome
- Idiopathic Hypercalciuria

Chronic kidney disease is a major cause of secondary hyperparathyroidism due to the inability to convert calcidiol to calciferol (active form of Vit D) in kidney. The low Vit D level results in reduced absorption of calcium by intestine, which leads to low serum calcium levels. Consequently, PTH is overstimulated (by low serum Ca) to compensate.

Bone

A living, active tissue made up of three main components:

- **Mineral:** mainly **calcium**
- **Matrix:** mainly **collagen fibers**
- **Cells:**
 - **osteoblasts** (bone-forming cells)
 - **osteoclasts** (bone-resorption cells)
 - **osteocytes** (derived from osteoblasts)

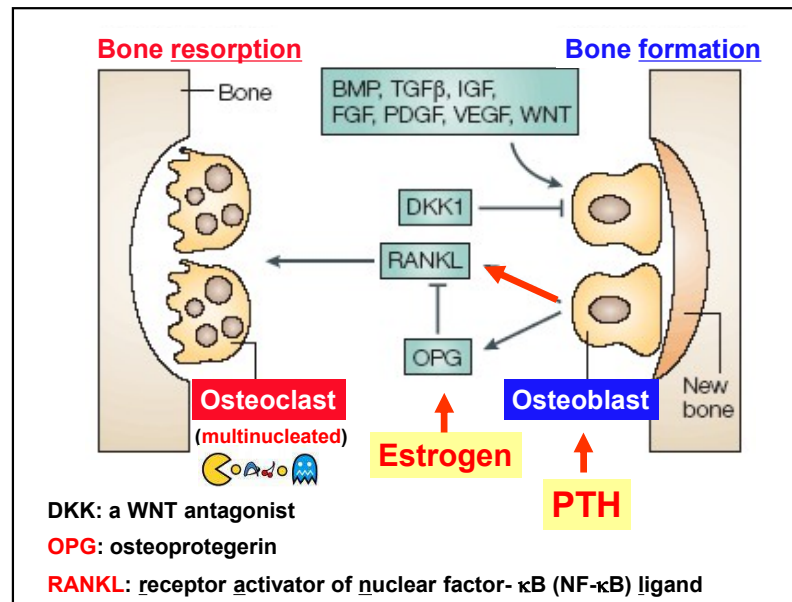
*** More than 70% of bone is hydroxyapatite**
 $\text{Ca}_5(\text{PO}_4)_3(\text{OH})$

Total body calcium- 1kg (99% in **bone**)

OSTEOBLASTS are the **cells that build new bone**. They have only one nucleus. Osteoblasts produce new bone called "**osteoid**" (made of bone collagen and other protein) and control calcium and mineral deposition. They are found on the surface of the new bone. Upon filling in a cavity, osteoblasts become flat which line the surface of the bone. These old osteoblasts are also called **LINING CELLS**. They respond to hormones by making special proteins (RANKL) that activate the osteoclasts.

OSTEOCLASTS are large **cells that chew the bone**. They are formed from two or more cells that fuse together, so the osteoclasts usually **have more than one nucleus**. They are found on the surface of the bone mineral next to the dissolving bone.

OSTEOCYTES are **cells inside the bone**. They also come from osteoblasts. Some of the osteoblasts turn into osteocytes while the new bone is being formed, and the osteocytes then get surrounded by new bone. They are not isolated, however, because they send out long branches that connect to the other osteocytes. These cells can sense pressures or cracks in the bone and help to direct where osteoclasts will dissolve the bone.



Bone is NOT an inert organ. During adult life, normal bone undergoes a continuous remodelling process of resorption and formation. This is normally a tightly coordinated process in which initial osteoclast resorption takes place in discrete packets, known as bone remodelling units, over a period of **about 8 days**. This is followed by a more prolonged phase of bone formation (**over about 3 months**) mediated by osteoblasts to repair the defect.

FYI

Wnt/ β -catenin pathway increases bone mass through a number of mechanisms including renewal of stem cells, stimulation of preosteoblast replication, induction of osteoblastogenesis, and inhibition of apoptosis of osteoblast and osteocyte.

Osteoprotegerin is a cytokine and a member of the TNF receptor superfamily. Osteoprotegerin production is stimulated by the estrogen, leading to **inhibition of RANKL**.

RANKL (Receptor Activator for Nuclear Factor κ B Ligand) activates osteoclasts.

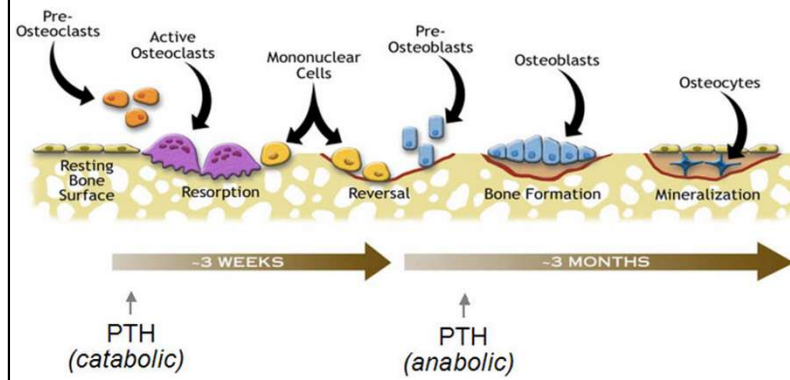
Overproduction of RANKL is implicated in a variety of degenerative bone diseases, such as RA.

Parathyroid Hormone

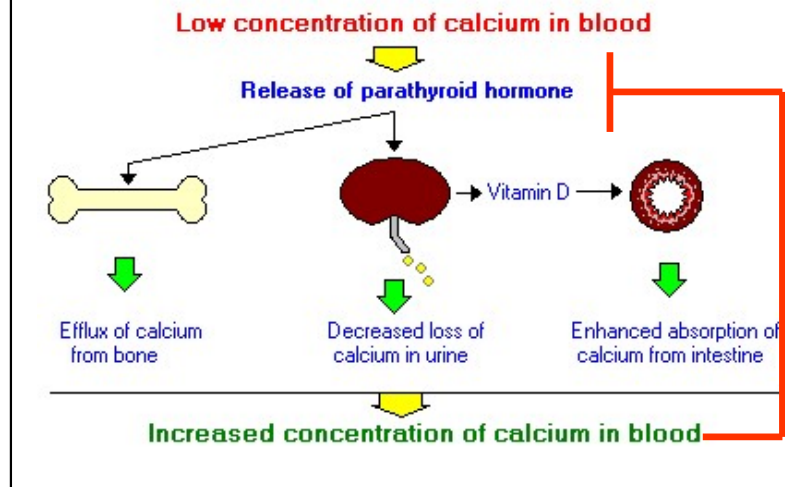
- Regulate **calcium** and **phosphorus** conc. in extracellular fluid
 - Mobilization of calcium from bone
 - ↑ **osteoclast differentiation and bone resorption** (via activation of **RANKL**)
 - ↑ **absorption** of calcium from the intestine
 - ↑ production of active Vit D in the kidney
 - ↓ **calcium loss in urine**
 - ↑ **reabsorption** of calcium in kidney
 - ↑ **loss of phosphate** in urine

PTH stimulates RANKL expression by osteoblasts, leading to osteoclast differentiation and the subsequent bone resorption.

Bone Remodeling Cycle



Parathyroid Hormone



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Vit D and Analogs



☐ Vitamin D & its major metabolites

- Vit D₃ (cholecalciferol)-D₃
- Vit D₂ (ergocalciferol)-D₂
- 25-Hydroxyvitamin D₃ (calcifediol)- 25(OH)D₃
- 1,25-Dihydroxyvitamin D₃ (calcitriol)- 1,25(OH)₂D₃
- 24,25-Dihydroxyvitamin D₃ (secalciferol)- 24,25(OH)₂D₃
- Dihydrotachysterol- DHT

☐ Vit D analogs

- Calcipotriene (calcipotriol)
- 1 α - Hydroxyvitamin D₂ (doxercalciferol)
- 19-nor-1,25-Dihydroxyvitamin D₂ (paricalcitol)

Vit D refers to two biologically inactive precursors - D₃, also known as cholecalciferol, and D₂, also known as ergocalciferol. The former is produced in the skin on exposure to UVB radiation (290 to 320 nm). The latter is derived from plants (UVB converts ergosterol to ergocalciferol) and only enters the body via the diet.

Both D₃ and D₂ precursors are hydroxylated in the liver and kidneys to form 25- hydroxyvitamin D (25(OH)D), the non-active 'storage' form, and 1,25-dihydroxyvitamin D (1,25(OH)₂D), the biologically active form that is tightly controlled by the body.

UVB converts 7-dehydrocholesterol into dihydrotachysterol. DHT undergo hydroxylation in the liver but no further hydroxylation in the kidney

Non-prescription Vit D: ergocalciferol & cholecalciferol

D₂ does not exist in high amount in humans.

Ergocalciferol is indicated for use in the treatment of hypoparathyroidism, refractory rickets, also known as vitamin D resistant rickets, and familial hypophosphatemia.

Calcifediol serum level reflects Vit D level in the body. Calcitriol is regulated by PTH and phosphate level so reduced calcitriol level is only seen in severe D deficiency.

Calcipotriol: topical use for psoriasis

Doxercalciferol & paricalcitol: treat secondary hyperparathyroidism in patients with kidney failure

Vitamin D & Analogs

- **Cholecalciferol (Vit D3)**
 - Naturally occurring form of Vit D; made by the **skin** upon **sun** exposure.
- **Calcidiol (25-hydroxyvitamin D3)**
 - **Pre-hormone** made in the **liver** directly from cholecalciferol
 - **Low** biologic activity, but is the **major** form in the circulation
 - Measured in typical **blood test** for Vit D deficiency
- **Calcitriol (1,25-dihydroxyvitamin D3)**
 - **Activated** form made from **calcidiol** in the **kidney**
- **Ergocalciferol (vitamin D2)**
 - From **plants** (not naturally occurring in the body)
 - Used in prescription vitamin D (Drisdol)

Calcitriol, the active form of vitamin D, has a half-life of about 15 hours, while calcidiol (aka calcifediol) has a half-life of about 15 days. Vitamin D binds to receptors located throughout the body.

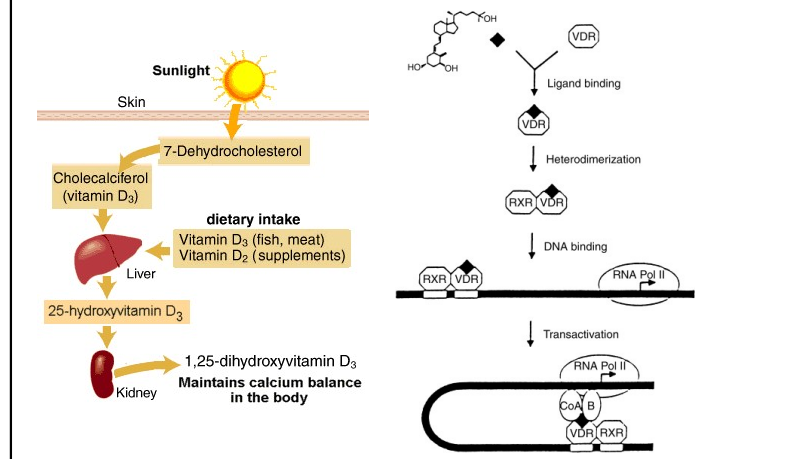
Vitamin D & Analogs

- ❑ Vitamin D acts by binding to **Vitamin D receptors (VDR)** located throughout the body
 - Regulate the **calcium** & **phosphorus** levels in the blood
 - **↑ absorption** in the **intestine**
 - **↑ reabsorption** in the **kidney**
 - **Inhibit PTH** secretion
 - Affects the immune system *via* gene regulation

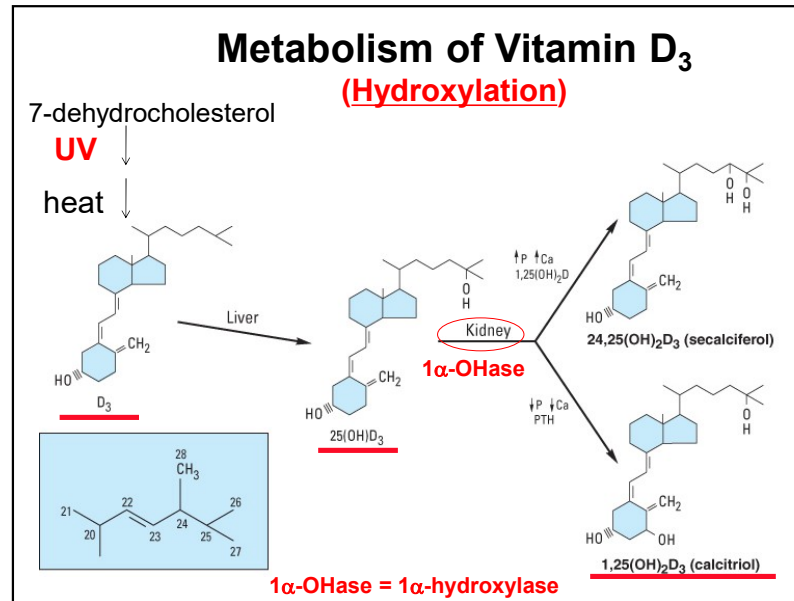
	Net effect on serum levels	
	Calcium	Phosphorus
PTH	↑	↓
Vit D	↑	↑

THE CLASSICAL FUNCTION OF vitamin D is to regulate mineral ion homeostasis. In addition to these effects, vitamin D mediates diverse cellular processes, including effects on cell growth and differentiation, on the immune system and on hormone secretion. 1,25-Dihydroxyvitamin D3 [1,25(OH)₂D₃] is the most active metabolite of vitamin D, and vitamin D receptors (VDRs) mediate its effects by affecting the transcription of 1,25(OH)₂D₃-responsive genes. Intestine, bone, and kidney are the three major target tissues involved in the regulation by 1,25(OH)₂D₃ of mineral homeostasis.

Vit D and Vit D Receptor (VDR) Mediate Gene Expression



Sunlight converts 7-dehydrocholesterol (7-DHC) in the skin to vitamin D₃, which is converted successively to 25-hydroxy-D₃ (25-D₃) and then to 1,25-dihydroxy-D₃ (1,25-D₃). Sunlight also induces expression of the vitamin D receptor (VDR). 1,25-D₃ and the VDR then together induce the expression of various gene.



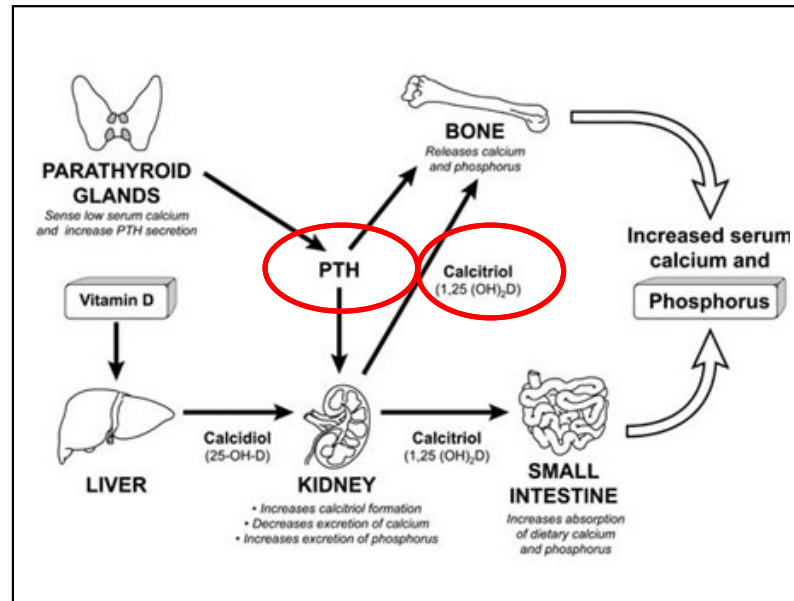
1,25-Dihydroxyvitamin D₃ [1,25(OH)₂D₃] is the most active metabolite of vitamin D.

1,25(OH)₂D₃ is produced by two sequential hydroxylations of vitamin D by 25-hydroxylase in the liver and by 25-hydroxyvitamin D₃ 1α-hydroxylase (1α-hydroxylase) in the kidney.

1,25(OH)₂D₃ can be further hydroxylated in the kidney by 25-hydroxyvitamin D₃ 24-hydroxylase (24OHase). 24-Hydroxylation of 1,25(OH)₂D₃ is believed to be the first step in the catabolism of the hormonally active form of vitamin D.

Measure 25(OH) Vit D, NOT 1,25(OH)₂ Vit D

1,25(OH)₂ vitamin D is more difficult and expensive to measure than 25(OH)D; moreover, it is not a good measure of vitamin D status. When patients are vitamin D deficient, the parathyroid hormone increases and drives the renal 1-α-hydroxylase, so that 1,25(OH)₂ vitamin D levels increase. Only in severe deficiency, when substrate is depleted, does the 1,25(OH)₂ vitamin D become low.



PTH up-regulates 25-hydroxyvitamin D3 **1-alpha-hydroxylase**, the enzyme responsible for 1-alpha hydroxylation of 25-hydroxy vitamin D, converting vitamin D to its active form (1,25-dihydroxy vitamin D). This activated form of vitamin D increases the absorption of calcium (as Ca^{2+} ions) by the intestine via calbindin.

Calbindin : vitamin D-dependent calcium binding proteins in intestine and kidney.

Calbindin-D9k is present in mammalian intestinal epithelial cells (enterocytes) and mediates the transport of calcium across the enterocytes from the apical side, where entry is regulated by the calcium channel, to the basolateral side, where calcium pumps utilize intracellular adenosine triphosphate to pump calcium into the blood. Calbindin-D9k can also be found in the kidney and uterus in some mammalian species.

The transport of calcium across the enterocyte cytoplasm appears to be rate-limiting for calcium absorption in the intestine; the presence of calbindin increases the amount of calcium crossing the cell without raising the free concentration.

Vit D, but NOT PTH, increases serum phosphate

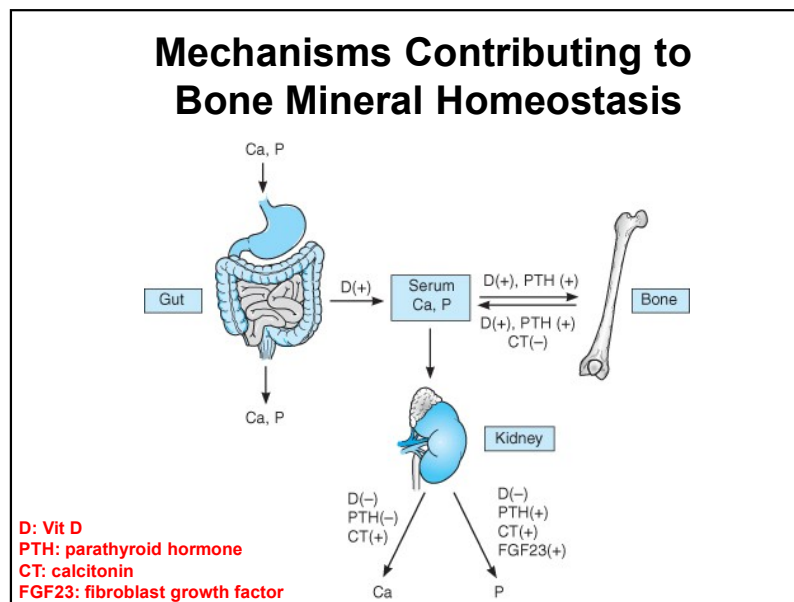
PTH promotes loss of phosphate in the urine (which forces dissolution of hydroxyapatite in the bone to release more phosphate which consequently release calcium also)

The mechanisms by which 1,25(OH)₂D regulates transcellular calcium transport are best understood in the intestine. Here 1,25(OH)₂D stimulates calcium entry across the brush border membrane into the cell, transport of calcium through the cell, and removal of calcium from the cell at the basolateral membrane. Calcium entry at the brush border membrane occurs down a steep electrochemical gradient. It is controlled by specific calcium channels called TRPV6 as well as by calmodulin bound to an intestinal specific myosin called brush border myosin 1. Transport of calcium through the cell is regulated by a class of calcium binding proteins called calbindins. Much of the transport occurs within vesicles that form in the terminal web. Removal of calcium from the cell at the basolateral membrane requires energy and is mediated by the ATP requiring calcium pump or CaATPase. 1,25(OH)₂D induces TRPV6, the calbindins, and the CaATPase, but not all aspects of transcellular calcium transport are a function of new protein synthesis.

Vitamin D Drug Interaction

- ❑ **Lipophilic**: intestinal absorption reduced by **statins** and **cholestyramine** (take the drug few hrs apart)
- ❑ **Corticosteroids blocks** Vit D- dependent calcium absorption in the intestine
- ❑ **p450 inducers** (**phenytoin** and **phenobarbital**) increase Vit D liver metabolism to inactive product

Cholestyramine is a bile acid sequestrant, which binds bile in the GI to prevent its reabsorption



Primary hormones that regulate bone minerals:

Vit D and PTH **promotes bone formation** (act on osteoblasts).

PTH paradoxically causes net bone loss (resorption) when administered in a continuous fashion, and net bone formation (deposition) when administered intermittently.

Vit D and PTH also **increases serum calcium** (when serum calcium is low) by stimulating osteoclast activity

Secondary hormone that regulate bone minerals:

Calcitonin (a **thyroid hormone**) **lowers** serum calcium by

inhibiting bone resorption and by **promoting calcium (and phosphate) loss** in the urine.

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Glucocorticoid: negative effect on bone health (long term use leads to osteoporosis)

Secondary Hormonal Regulators of Bone Mineral Homeostasis

❑ Calcitonin

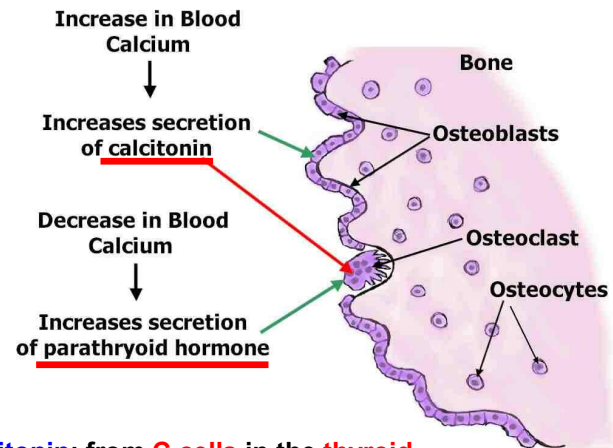
- Secreted by the **parafollicular cells** of the thyroid
- Stimulated by **hypercalcemia**
- ↓ serum **calcium** and **phosphate**
 - ↓ **bone resorption** by **osteoclasts**
 - ↓ **reabsorption** of calcium & phosphorus in **kidney**
- Clinical uses: treat **Paget's disease of the bone**, **hypercalcemia**, and **osteoporosis**
- Side effects: GI symptoms, nausea, flush



parafollicular cell is found as single cells in the epithelial lining of the follicle or in groups in the connective tissue between follicles in the thyroid gland. They usually appear as large, clear cells since they do not stain well with hematoxylin and eosin. They are sometimes called **parafollicular cells** based on their location and **clear cells (C cells)** based on their appearance of their cytoplasm.

Parafollicular cells secrete calcitonin, a hormone that lowers the level of calcium in the blood.

Calcitonin *tones down* serum calcium



Calcitonin: from **C cells** in the **thyroid**

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Glucocorticoids

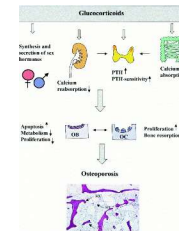
❑ MOA (on bone metabolism)

- ↓ Vit D-mediated calcium **absorption**
 - ↑ renal calcium **excretion**
 - ↓ bone **formation**
 - ↑ osteoclast **differentiation**
- ↓ serum calcium
- osteoporosis

❑ Clinical uses (e.g, **prednisone**, **dexamethasone**)

- Treat **hypercalcemia** associated with malignancy & granulomatous disease
- Useful in **Vit D intoxication**

❑ **Prolonged use of glucocorticoids is a common cause of osteoporosis**



Vit D promotes calcium transport through the gut wall

e.g., lymphoma-associated hypercalcemia & **sarcoidosis**-increased $1,25[\text{OH}]_2\text{D}_3$

Important Drug-induced Osteoporosis

- ❑ Glucocorticoids
- ❑ Heparin
- ❑ Antiepileptic agents (phenytoin, carbamazepine)

** **Glucocorticoid-induced osteoporosis (GIO)** is the 2nd most prevalent cause of osteoporosis after **postmenopausal estrogen deficiency**.*

T score: compared with young normal reference

normal range: -1.0 or higher; osteopenia: -1.0 and -2.5

Osteoporosis: -2.5 or lower

Z score: compared to the age-matched normal (used in cases of severe osteoporosis in patients <50 yo)

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Estrogens & SERMs

- ❑ **Estrogens** (e.g., estradiol)
 - ↓ **PTH-mediated bone-resorption** (via **OPG**)
 - **Clinical use**
 - Treat/prevent **postmenopausal osteoporosis**
 - **Long term use is discouraged due to side effects on breast, uterus & CVS**
- ❑ **Selective estrogen receptor modifier (SERMs)**
 - **Differential activity on ER** (tissue-dependent)
 - **Raloxifen** (**ER agonist on bone**)
 - Prevent spine but *not* hip fractures

PTH stimulates RANKL expression by osteoblasts, leading to osteoclast differentiation and the subsequent bone resorption.

Estrogen induces expression of OPG which inhibits RANKL

Long term use of estrogens promotes development of breast cancer and endometrial carcinoma.

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Recombinant PTH

Teriparatide (PTH 1-34), Abaloparatide (PTHrP 1-34)

- ☐ Intermittent use promotes bone formation more than bone resorption (stimulate osteoblasts)
- ☐ SC injections (once daily for up to 2 years) 
- ☐ Treat/prevent **glucocorticoid-induced** or **postmenopausal osteoporosis** or **men with osteoporosis** (often follow up with a **BP**)
- ☐ Side effects: hypotension (dizziness, nausea)
- ☐ **US box warning:** risk of **osteosarcoma** (dose-duration dependent) in animal studies



Abaloparatide is a homolog of PTH related protein (PTHrP).

Both teriparatide and abaloparatide are anabolic agents (stim bone formation)

After two years the effectiveness stalls and risks increase

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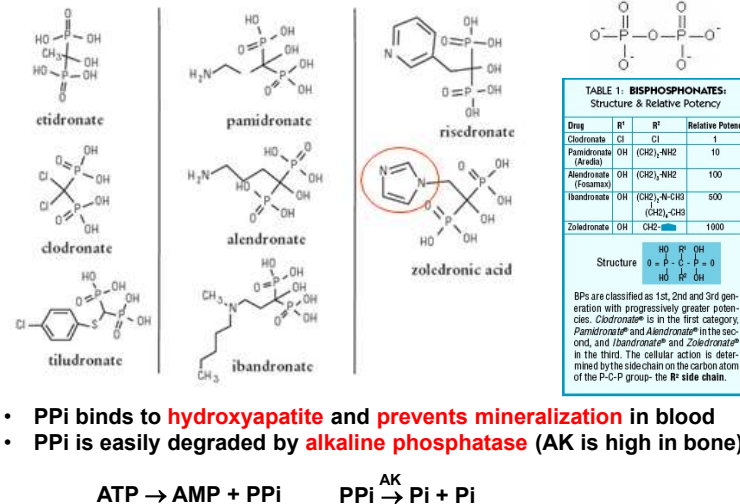
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Bisphosphonates are **PPi** analogs

Agents used to treat osteoporosis



Bisphosphonates are stable analogues of pyrophosphate (PPi).

PP_i is important in calcium metabolism because **PP_i can prevent the precipitation of calcium phosphates (esp in the blood), and can slow the rates at which hydroxyapatite crystals grow and dissolve**. It is degraded by alkaline phosphatase in bone structures.

Pyrophosphate has a P-O-P structure, whereby two phosphate groups are linked by an oxygen atom. Bisphosphonates, however, have a P-C-P structure, a central carbon atom replacing the oxygen. The P-C-P structure makes BPs less degradable by AP in the bones.

Side chains R₁ and R₂ are attached to the carbon atom and influence the bisphosphonates' ability to bind to bone, and their anti-resorptive ability. Bisphosphonates containing a primary nitrogen atom in the R₂ side-chain (e.g., pamidronate) are more potent than non-nitrogen bisphosphonates, (e.g., clodronate), whilst modifying the primary amine to form a tertiary amine (e.g., ibandronate) results in even more potent molecules. **The most potent bisphosphonates to date appear to be those containing a tertiary amine within a ring structure, such as zoledronic acid.**

Bisphosphonates bind avidly to hydroxyapatite bone mineral surfaces and are selectively internalised by osteoclasts where they inhibit their activity.

Nitrogen-containing bisphosphonates inhibit the mevalonate pathway, the main target being farnesyl diphosphate (FPP) synthase.

Inhibition of the mevalonate pathway leads to loss of important prenylated proteins which are required for post-translation lipid modification (prenylation) of signalling GTPases, such as Ras, Rho and Rac.

These regulate a variety of key osteoclast cell functions, **and loss of these proteins leads to induction of osteoclast apoptosis.**

Non nitrogen-containing bisphosphonates have a different mechanism of action. They are metabolically **incorporated into nonhydrolysable analogues of ATP**, ultimately also leading to osteoclast apoptosis. As a consequence of this osteolysis is effectively inhibited.

Bisphosphonates

Potency: C<E<P<A<I<R<Z

$$\begin{array}{c}
 \text{O} & & \text{R}_1 & & \text{O} \\
 \parallel & & | & & \parallel \\
 \text{O} - \text{P} - & \text{C} & - \text{P} - \text{O} \\
 | & & | & & | \\
 \text{O}^- & & \text{R}_2 & & \text{O}^-
 \end{array}$$

$$\begin{array}{c}
 \text{O} & & \text{O} \\
 \parallel & & \parallel \\
 \text{O}^- - \text{P} - \text{O} - \text{P} - \text{O}^- \\
 | & & | \\
 \text{O}^- & & \text{O}^-
 \end{array}$$

MOA

- **Analogs of pyrophosphate (PPi)***
 - BPs (P-C-P core) is *less degradable* than PPi
- **Inhibit bone resorption**
 - Bind to **hydroxylapatites**
 - Promotes **osteoclast apoptosis**
 - Modest **↑ in bone mineral density**

Agents used to treat osteoporosis	
1 st	clodronate
	etidronate
2 nd	pamidronate
	alendronate
3 rd	ibandronate
	riseudronate
	zoledronate

BP binding to bone → BP uptake by OCs → OCs apoptosis

$$\left[\begin{array}{c} \text{O} & \text{O} & \text{O} \\ \parallel & \parallel & \parallel \\ \text{O} - \text{P} - \text{O} - \text{P} - \text{O}^- \\ | & & | \\ \text{O}^- & & \text{O}^- \end{array} \right] \left[\text{Ca}^{2+} \right]_2$$

*PPi inhibits calcification in plasma but is easily degraded by alkaline phosphatase in bone

The development of the bisphosphonates for the treatment of skeletal disorders associated with increased bone resorption has been a major advance in the management of Paget's disease. These drugs, initially known as diphosphonates, are analogues of inorganic pyrophosphate, a factor believed to be a necessary component for the mineralization of bone. Pyrophosphate is a natural circulating inhibitor of mineralization in the blood and urine which can't get inside the bones because the lining cells destroy it with alkaline phosphatase.

All bisphosphonates have a central P-C-P core, which was substituted for the naturally occurring P-O-P core of pyrophosphate, because unlike P-O-P, the P-C-P structure is impervious to metabolic degradation. The bisphosphonates bind to hydroxyapatite. The primary effect of bisphosphonates is to inhibit osteoclastic bone resorption.

As the bone remodeling cycle is slowed, formation and resorption are decreased, although resorption is inhibited more than formation. This process leads to a modest increase in bone mineral density. The osteogenic action of bisphosphonates and their relative lack of activity outside the skeleton have led to the increased use of these agents in treating osteoporosis.

Approved for treatment/prevention of osteoporosis:

alendronate, ibandronate, risedronate & zoledronate

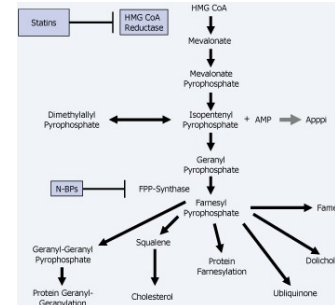
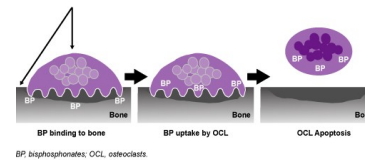
Zoledronate is used IV once yearly for treatment or once every two yr for prevention

Bisphosphonate

Agents used to treat osteoporosis

□ MOA (continued)

- **Amino-bisphosphonates (N-BPs)** are more potent
- **N-BPs blocks farnesyl pyrophosphate (FPP) synthase** in the mevalonate pathway, leading to **inhibition of osteoclast attachment** and thus **reduced bone resorption**



The earliest bisphosphonates (e.g., etidronate), appear to achieve their effects by generating nonhydrolyzable analogues of ATP, while the later generation of more potent aminobisphosphonates, such as pamidronate and risedronate, inhibit protein prenylation through inhibition of farnesyl pyrophosphate synthase, a key enzyme in the mevalonate pathway

Inhibition of the mevalonate pathway leads to loss of important prenylated proteins which are required for post-translation lipid modification (prenylation) of signalling GTPases, such as Ras, Rho and Rac.

These regulate a variety of key osteoclast cell functions, and loss of these proteins leads to induction of osteoclast apoptosis.

* **Statins** may also *stimulate* bone formation by

blocking mevalonate synthesis

Non nitrogen-containing bisphosphonates have a different mechanism of action. They are metabolically incorporated into nonhydrolysable analogues of ATP, ultimately also leading to osteoclast apoptosis.

Bisphosphonate

Agents used to treat osteoporosis

- ☐ **Poor GI absorption**
 - Preferred oral drugs: **alendronate**, **risedronate**
 - Preferred IV drug: **zolendronate** (once/yearly)
- ☐ **50% of the absorbed drugs taken up by bones**
- ☐ **Absorbed drugs accumulate in bone for months**
- ☐ **Eliminated in urine (unaltered)**
- ☐ **Clinical uses:**
 - **Paget's disease of the bone**
 - **Postmenopausal osteoporosis**
 - **Hypercalcemia associated with malignancy**



IV BPs (e.g., zolendronate) do not cause gastric & esophageal irritation

Paget's disease, characterized by excessive and abnormal remodeling of bone, is a common disorder in middle-aged and elderly patients. The excessive remodeling gives rise to bones that are extensively vascularized, weak, enlarged, and deformed, with subsequent complications.

Pathophysiology of Paget's disease : bone destruction followed by disorganized bone formation

Excessive osteoclastic activity (resorption of normal bone by giant, multinucleated cells) begins the cycle. Subsequently, an intense osteoblastic response produces increased disorganized bone formation and connective tissue reaction. As the osteoclastic and osteoblastic activities of bone destruction and formation repeat, a high degree of bone turnover occurs.

The etiology of Paget's disease is unknown. Genetic and nongenetic factors have been implicated in the pathogenesis of this disease.

Bisphosphonates

Agents used to treat osteoporosis

□ Side effect:

- **Gastric & esophageal irritation** (oral BP)
**to minimize: take it with 8 oz water first thing in the morning and remain upright for >30 min!!*
- **Risk of osteonecrosis of the jaw** (consult about planned dental procedure and risk)
- **Hypocalcemia**
- **Atypical femoral fractures** (subtrochanteric or diaphyseal fractures)
- **Increased toxicity in patients with renal disorder**



Regular fragility fractures: spine, hip, wrist, rib

Atypical femoral fracture; median >7 yrs

ONJ risk factors: IV, cancer.

Hormones and Drugs that Affect Bone Mineral Homeostasis

Primary hormonal regulator	Parathyroid Hormone (PTH)
	Vit D & analogs
Secondary hormonal regulator	Calcitonin
	Glucocorticoids
	Estrogens
Drugs	Raloxifen (SERM)
	Teriparatide (PTH 1-34), Abaloparatide
	Bisphosphonates (BPs)
	<u>Denosumab</u>
	Romsozumab

Primary hormones that regulate bone minerals: PTH & Vit D

Secondary hormones that regulate bone minerals: calcitonin, glucocorticoids, estrogens

Drugs that are used to treat bone mineral disorders: BPs, PTH analog, SERM, calcimimetics (cinacalcet)

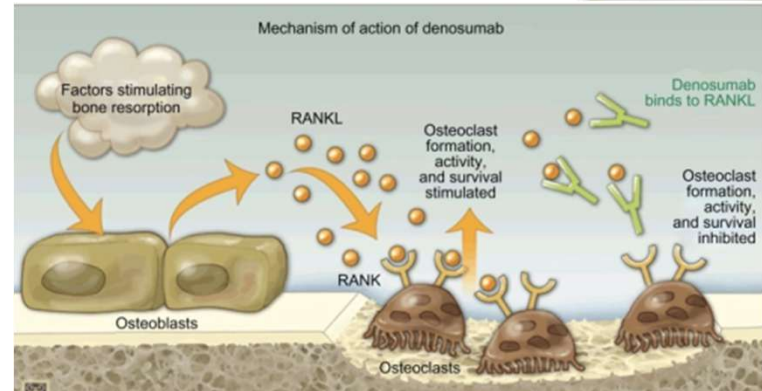
Calcimimetics (cinacalcet) is mainly used to reduce PTH in patients treat hyperparathyroidism (will be discussed in IS-GERD)

Glucocorticoid: negative effect on bone health (long term use leads to osteoporosis)

Anti-RANKL mAb

Denosumab

❑ MOA: a **RANKL inhibitor**



RANK (receptor activator of nuclear factor kappa) is a TNF-alpha family protein expressed on the surface of pre-osteoclasts. Upon activation by RANKL (released by osteoblasts), osteoclasts differentiation and maturation proceed, leading to bone resorption and increased calcium conc in blood.

Anti-RANKL mAb Denosumab

- ❑ Clinical use
 - Prevent postmenopausal osteoporosis at high risk for fractures (not as initial therapy), bone metastasis & giant cell tumor of the bone.
 - Correct hypocalcemia or Vit D deficiency first
- ❑ Side effects
 - Joint and muscle pain in the arms and legs
 - Increased risk of infections
 - Risk of osteonecrosis of the jaw (ONJ)

Contraindicated in hypocalcemia

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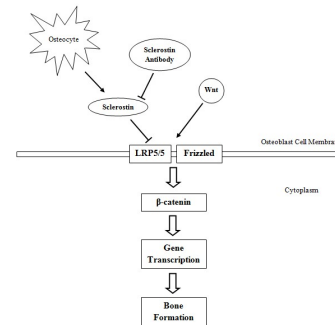
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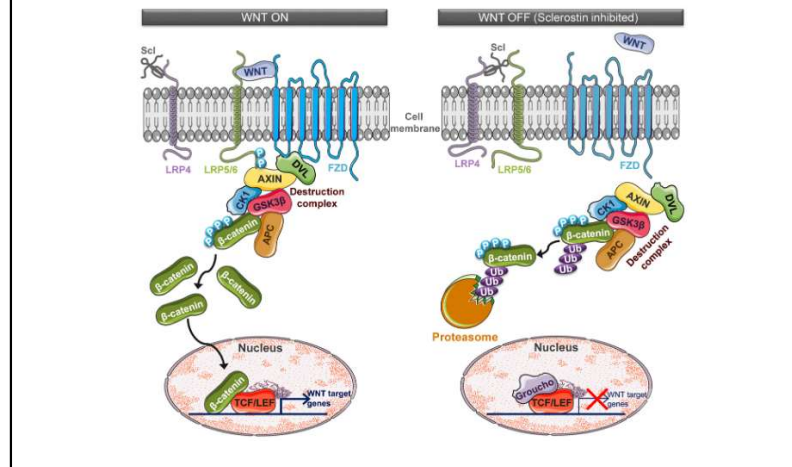
Sclerostin as a therapeutic target for bone diseases

- A glycoprotein secreted mainly by **osteocytes**
- Serum level increases with age
- **Inhibits Wnt signaling leading to inhibition of bone formation**



Modulation of Wnt signaling by sclerostin. The binding of Wnt proteins to the LRP5/6 receptor and frizzled co-receptor on osteoblasts leads to stabilization of beta-catenin and regulation of gene transcription that promotes osteoblast bone formation. Sclerostin is secreted by osteocytes and can antagonize Wnt signaling and inhibits bone formation. Monoclonal antibodies that block sclerostin enhance Wnt signaling and increase osteoblastic bone formation.

Sclerostin as a therapeutic target for bone diseases



A simplified overview of the canonical WNT signalling pathway, and its inhibition by sclerostin. In the WNT ON state, extracellular WNT ligands bind the LRP5/6 cell surface receptor and a frizzled (FZD) family cell-surface receptor to form a ternary complex, leading to translocation of the destruction complex to the cell membrane, phosphorylation of the intracellular region of LRP5/6 and recruitment of AXIN which disrupts destruction complex activity. Non-phosphorylated β -catenin accumulates in the cytoplasm and translocates to the nucleus where it binds the TCF/LEF transcription factors to initiate transcription of *WNT* target genes. In the WNT OFF (sclerostin inhibited) state, sclerostin (Scl) binds to the LRP4 and LRP5/6 cell-surface receptors, preventing the interaction between WNT ligands with LRP5/6. This prevents formation of the LRP5/6–WNT–FZD ternary complex and hence inhibits canonical WNT signalling. Cytosolic β -catenin is phosphorylated and targeted for ubiquitination by the destruction complex and degraded in the proteasome. TCF/LEF transcription factors in the nucleus are associated with Groucho and *WNT* target gene transcription is repressed. This figure was produced using Servier Medical Art.

Sclerostin Inhibitor

Romosozumab

- ❑ **MOA:** Increases bone formation and decrease bone resorption by binding to and inhibiting sclerostin
- ❑ **Clinical use** (2 injs sequentially once a month for a year)
 - For women with a history of osteoporetic fracture or multiple risk for fractures (not as initial therapy).
 - Followed by an antiresorptive agent (e.g., BP)
- ❑ **Boxed warning**
 - Increased risk for **myocardial infarction, stroke, and CV death**



avoided in patients with recent CV events

Pharmacologic Therapy for Osteoporosis

Candidates:

- Postmenopausal women and men ≥ 50 yo with a history of fragility fracture (hip or vertebral) or with osteoporosis (**BMD T ≤ -2.5**)
- Postmenopausal women who are at high risk for fracture (osteopenia; $-2.5 \leq T \leq -1$)

Pretreatment Eval: Calcium, 25[OH]D, creatinine

- Correction of **Ca** and **25[OH]D** is required for oral/IV BP
- Discuss risk of **ONJ** (inquire about invasive dental procedure)

Choice of Drug:

- Based on efficacy, safety, cost, convenience, patient factors
- Should have **normal serum calcium** and **25-hydroxyvitamin D**
- Should receive **supplemental** calcium and Vit D if dietary intake is **inadequate**
- Initial therapy: **oral BP (alendronate or risedronate)**
 - Contraindications: patients with **esophageal disorders** or **difficulty in dosing requirement** or **CKD**
 - Alternative: **IV BP or denosumab, teriparatide, SERMs**

Fracture risk assessment: a combination of BMD and clinical risk factors

FRAX (<https://www.sheffield.ac.uk/FRAX/>): estimate the 10-yr probability of hip fracture or major osteoporotic fractures combined (hip, spine, shoulder, or wrist) or an untreated patient using femoral neck BMD and easily obtained clinical risk factors for fracture

Treatments for Osteoporosis

- ❑ **Estrogen**
 - for **postmenopausal** osteoporosis
 - Cycled with a progestin to prevent endometrial carcinoma
- ❑ **Selective estrogen receptor modulators (SERMs)**
 - **Raloxifen**: protects **spine fracture** (not hip)
 - Use **Vit D suppl.** to counter the reduced intestinal calcium transport associated with osteoporosis
- ❑ **Teriparatide (PTH 1-34), abalaparotide (PTHrP 1-34): anabolic**
- ❑ **Calcitonin:**
 - Inhibits bone resorption by osteoclasts
- ❑ **Bisphosphonates**
 - Inhibits bone resorption by osteoclasts
- ❑ **Denosumab:**
 - Inhibits osteoclast maturation by binding to **inhibiting RANKL**
- ❑ **Romosozumab:**
 - **Inhibits sclerostin** leading to enhanced bone formation

Other Disorders of Bone Mineral Homeostasis



❑ Paget's Disease of Bone

- A localized bone disease characterized by *uncontrolled* osteoclastic bone resorption with secondary increase in abnormal bone formation
- Elevated serum **alkaline phosphatase** & urinary **hydroxyproline** (diagnosis)
- **Calcitonin** and **bisphosphonates** are **1st-line agents** (< 6 months); **plicamycin** may be used when other less toxic agents have failed

Paget's disease : a chronic bone disorder that is due to irregular breakdown and formation of bone tissue.

Paget's disease can result in bone pain, arthritis, bone deformity and fractures.

The actual cause of Paget's disease is not known.

kidney stones: calcium oxalate, calcium phosphate, mix, uric acid, cystine, magnesium ammonium phosphate (infection stone)

Calcium oxalate develops in acid urine. Calcium phosphate develops in alkaline.

Genetically inherited disorders account for some stone formation. High protein and salt intake increase the risk of calcium stone formation. High purine diets (meat, fish, chicken) lower urinary pH and cause increased excretion of uric acid. Vitamin B6 deficiency leads to increased formation and excretion of oxalate. Dehydration, excessive vitamin C intake, calcium supplementation, and

calcium containing antacids may also lead to stone formation. Also, geography plays a part with more stones noted in the southeast, also known as the stone belt.



- ❑ The treatment of osteoporosis consists of **lifestyle measures** and **pharmacologic therapy**.
- ❑ **Lifestyle measures include**
 - adequate calcium and vitamin D
 - exercise
 - smoking cessation
 - counseling on fall prevention
 - avoidance of heavy alcohol use